

Pillar 5: BigQuery Data Export & Cloud Warehousing

Raw Telemetry Streaming, GCP Export Configuration & Enterprise SQL Warehousing | Apple Omni-Channel Store

PROPERTY NAME

Apple Omni-Channel Store

MEASUREMENT ID

G-X3R766JV6K

INTEGRATION TARGET

Google BigQuery (GCP)

EXPORT MODES

Daily Batch + Intraday Streaming

01 Context & Scope

The narrative



WHY LEAVE THE GA4 WEB UI

The standard GA4 web interface applies data sampling on large explorations, thresholding on small audiences, cardinality limits on high-variety dimensions, and a fixed retention window. Every one of those constraints disappears once raw event logs are streamed directly into BigQuery.

The GA4 UI is built for fast, approximate answers at the interface level, not for arbitrary long-horizon SQL analysis. Piping raw, unsampled event logs into BigQuery hands that constraint back to engineering: every event, every parameter, and every user property is queryable exactly as collected, with no rounding, no suppressed low-volume rows, and no 14-month ceiling on how far back a query can reach.

02 Warehouse & Export Configuration Specs

Batch vs. streaming, and where the data lives



Daily Export

BATCH · COST-EFFECTIVE

Delivers one partitioned table per day, ideal for scheduled analysis, historical modeling, and cost-sensitive pipelines that don't need same-hour visibility into new events.



Streaming Export

INTRADAY · LOW-LATENCY

Delivers events into an intraday table within minutes of collection, supporting operational dashboards and fraud or anomaly detection that can't wait for the next daily partition to land.

Both modes are configured under **GA4 Admin → Product Links → BigQuery Links**, where the property is linked to a specific Google Cloud project and dataset location is chosen explicitly, EU or US, since GA4 does not move data across that boundary automatically. That regionalization choice is a data-privacy decision as much as a technical one: it determines which jurisdiction's data-residency rules govern the exported tables from day one, and it can't be changed retroactively without re-exporting.

03 Advanced Enterprise Capabilities & SQL Use Cases

PORTFOLIO DEMO - NOT CLIENT WORK

What raw export actually unlocks downstream



Multi-Touch Attribution
Unnested JSON event params



Customer LTV Modeling
SQL cohort & revenue models



CRM Data Joining
Cross-channel identity joins



Looker Studio / Power BI
Direct BI tool integration



Vertex AI Readiness
ML pipeline & feature store input

The unnested, repeated JSON structure BigQuery preserves for event parameters and item arrays is what makes multi-touch attribution and LTV modeling tractable in SQL: a single query can flatten every touchpoint and every line item across the full customer history, then join that against CRM records by hashed email or Client ID for a genuinely cross-channel view, something the aggregated GA4 UI reports were never designed to expose.

04 Implementation Proof

Visual asset from the demo property

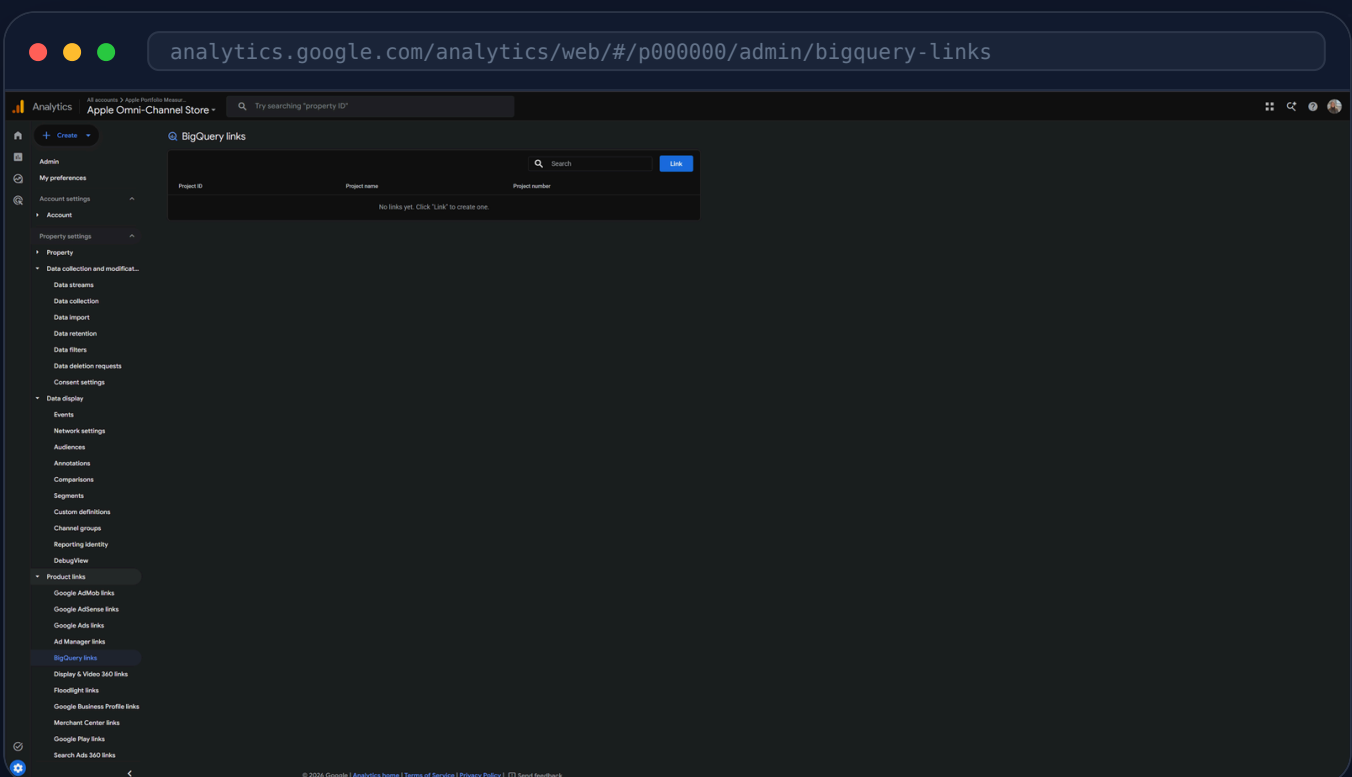


FIG. 6 GA4 BigQuery Links Configuration — confirms the BigQuery Links panel under Product Links is configured and ready, the exact integration surface documented in Section 2 for connecting this property to a Google Cloud project. No link has been created yet in this demo environment, which is expected: the panel shown is the setup surface itself, not a live production connection.